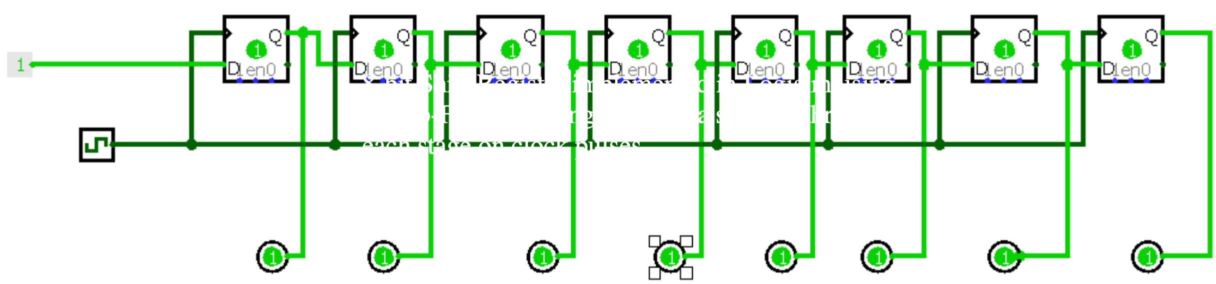
Experiment 4

Aim :
To design and simulate an 8-bit shift register using flip-flops in Logisim and observe the shifting of
data.using 8 × D Flip-Flops, Clock, Input pin (for serial data), LED Output pins (for Q0–Q7)Wiring tools.

Procedure :
An **8-bit shift register** is a sequential digital circuit made using 8 flip-flops connected in series. It stores binary data and shifts it one position to the next flip-flop on every clock pulse. In Logisim, this circuit is simulated virtually to observe how data moves step-by-step from input to output. It is commonly used for data storage, transfer, and serial communication.
First, open Logisim and create a new circuit project. From the memory components section, place eight D flip-flops on the workspace and arrange them in a straight line. Then, add a clock component and connect its output to the clock input of all the flip-flops so that they operate simultaneously. After that, place an input pin to provide the serial data and connect it to the input (D) of the first flip-flop. Next, connect the output (Q) of each flip-flop to the input (D) of the next flip-flop in sequence, forming a chain. Attach LEDs or output pins to each flip-flop's output to observe the data. Once all connections are complete, enable the clock from the simulation menu and start applying input values. As the clock pulses are generated, observe that the data shifts from one flip-flop to the next with each pulse



8-bit Shift Register implemented in Logisim using D Flip-Flops showing serial data shifting through each stage on clock pulses.

Results :

The 8-bit shift register was successfully designed, implemented, and simulated using Logisim. The circuit was constructed using eight D flip-flops connected in series, with a common clock signal applied to all stages. During the simulation, serial input data was provided to the first flip-flop, and the outputs were observed using LEDs connected to each stage.

It was clearly observed that with every clock pulse, the input data shifted sequentially from one flip-flop to the next. The bit entered at the first stage and propagated step-by-step through all eight stages until it reached the final output. This confirmed that the circuit operates correctly as a Serial-In Serial-Out (SISO) shift register